Gue Yi Peng

Singapore
Contact via portfolio site

EDUCATION

Temasek Polytechnic *04/2023 – 04/2026*

Diploma in Electronics

Relevant Coursework and Academic Projects

- **Embedded Systems & Automation: TrackSafe (RailTech Project)**
  - Designed a LoRa-based tracking system for MRT maintenance with a cross-disciplinary team
  - Cut safety incidents by ~40%
- **Software & Cloud Systems: Metabots Singapore Knowledge Platform**
  - Built and deployed a data pipeline using AWS (RDS, EC2, Lambda, Bedrock, Lex, Glue)
  - Cut annual cloud costs by over USD 2,000 and earned a feature at Interclean Amsterdam
- **Internet of Things: Ageless Bicycle Singapore Project**
  - Developed a low-cost BLE and RFID bike tracking system with real-time Blynk alerts
  - Improved bike recovery rate and reduce bike stealing rate

Institute of Technical Education *04/2020 – 04/2022*

Higher Nitec in Electronics Engineering

Relevant Coursework and Academic Projects

- **Embedded Control & Automation: Smart Trolley**
  - Built a LoRa and GPS asset-tracking system with Google Maps API integration
  - Cut asset loss by 40–60%
- **Embedded Systems: COVID-19 Digital Lanyard (SICK Project)**
  - Designed a wearable health screening device using an ESP32-CAM and thermal sensor
  - Achieved ~92% fever-detection accuracy
- **IoT & Home Automation: Smart Home System**
  - Integrated a Zigbee-based smart home system using a Raspberry Pi, WISE module and multiple sensors
  - Enabled sensor data logging on ThingSpeak and automated Twitter alerts

AWARDS

- Director's List, academic year (2021/2022, 2022/2023)
- Edusave Skills Award (2023, 2025)
- Edusave EAGLES Award (2022)
- Edusave Academic Achievement (2022, 2023)
- Service Star Award (Gold); ITE

WORK EXPERIENCE

SSMC (Systems on Silicon Manufacturing Company) — Product & Testing Engineer (Intern)

04/2025 – 08/2025

- Performed semiconductor yield analysis and process troubleshooting to resolve daily production issues
- Cleaned and managed data integrity using VBA and microsoft excel
- Conducted semiconductor reliability and process control testing on NXP

PORTFOLIOS, PROFILES

- linkedin.com/in/gue-yi-peng-0013491ba
- Google Drive portfolio folder
- Google Drive Amsterdam RAI

SKILLS

- **Programming:** Python, C, C++, JSON
- **Embedded Systems & IoT:** Arduino, Raspberry Pi, PIC, ESP32, LTspice, EAGLE, MPLAB
- **Data Tools:** Tableau, KNIME, MySQL, Applied Data Engineering (AWS Bedrock, Lex)
- **Platforms:** MATLAB, AWS, Ubuntu, Blynk, SketchUp

ACHIEVEMENTS

4 October 2025

IdeateCOMM Challenge 2025 Competition (Silver & Best Prototype)

30 November 2024

Singapore RailTech Grand Challenge 2024 (IHL Category, Bronze)

3 October 2024

Student Poster Competition, 2nd Place — SHEC (Singapore Healthcare Engineering Conference)

28 October 2023

Singapore Energy Grand Challenge (Youth), 1st Runner Up

25 May 2023

Lee Kuan Yew Technology Award (ITE) ; Smart Trolley

9 May 2023

Outstanding Graduate Award (ITE) ; Electronics Engineering

24 Jan 2022

The Sixth Sense Competition 6th edition (SILVER); SICK Sensor Intelligence

automotive chips
- Handled RMA wafers, including collection and scraping processes

SICK Sensor Intelligence — Test Engineer (Intern)
12/2022 – 03/2023

- Applied process and quality control via stress, heat and reliability testing on photoelectric and inductive proximity sensors
- Assembled and soldered prototypes for new photoelectric sensor designs
- Evaluated refractive index and false trigger performance under extreme conditions
- Performed temperature testing on reed switches to assess operational reliability

CO-CURRICULAR ACTIVITIES

October 2023 – June 2024
Electronics Interest Group (Outreach Leader)

- Coordinated Electronic orientation to welcome the Year 1s, collaborating with team members and planning suitable activities for bonding sessions
- Coached secondary school students from Tampines Secondary to build an electronic prototype from scratch, teaching them soldering, 3D printing and coding
- Arranged and set up an exhibition at Bedok Green Secondary School to promote and represent Temasek Polytechnic's Electronic course
- Presented at Homecoming event 2023 for alumni

4 February 2026
Engineering Tech Show (Metabots)

- Showcased the completed knowledge-based website to industry professionals
- Collaborated with Metabots Singapore, promoting other products developed by Metabots Singapore
- Networked with potential industry partners interested in collaborating with Metabots Singapore

REFERENCES

- Available on request